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Exercise Sheet 6

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Compute the following remainders

1. $R_{11}(24^{100} \cdot 33^{50})$.
2. $R_7(8^{2025} + 14^{2026})$.
3. $R_5(12^{2026} \cdot 3^{100})$.
4. $R_9(18^{100} + 28^{50})$.

Exercise 2.
Prove that $13 \mid 2^{70} + 3^{70}$.

Exercise 3.
Let $n \in \mathbb{N} \setminus \{0\}$. Prove that $n^2 \mid (n+1)^n - 1$.

Exercise 4.
Let $k \in \mathbb{N}$. Prove that $19 \mid 2^{2^{6k+2}} + 3$.

Exercise 5.
Let $a, b \in \mathbb{Z}$ with $b > 0$. Suppose a can be expressed in the following ways using the division algorithm

$$a = bq_1 + r_1 \quad a = bq_2 + r_2$$

for $0 \leq r_1 < b$ and $0 \leq r_2 < b$.

Prove that $q_1 = q_2$ and $r_1 = r_2$.

Exercise 6.
Find all solutions to the following system of congruences

$$\begin{aligned} 3x + 4y &\equiv 5 \pmod{13} \\ 2x + 5y &\equiv 7 \pmod{13} \end{aligned}$$

Exercise 7.
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^3 - 3y^3 = 9$$

Exercise 8.
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^4 - 5y^2 = 2$$

★ **Exercise 9.**
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^3 - 3 = y^2$$

Hint: For a prime p such that $p \nmid a$ we have $a^{p-1} \equiv 1 \pmod{p}$

★ **Exercise 10.**
Prove that $\gcd(n, 2^{2^n} + 1) = 1$ for $n = 1, 2, \ldots$